Augmenting phenotype prediction models leveraging a genomic Large Language Model

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Introduction

Motivation

Huntington's Disease (HD)

- Inherited neurodegenerative disorder
- Leads to progressive brain cell damage and loss of function
- No cure

Motivation

How can integrating genomic language models improve our understanding of HD and help the process of finding new therapies?

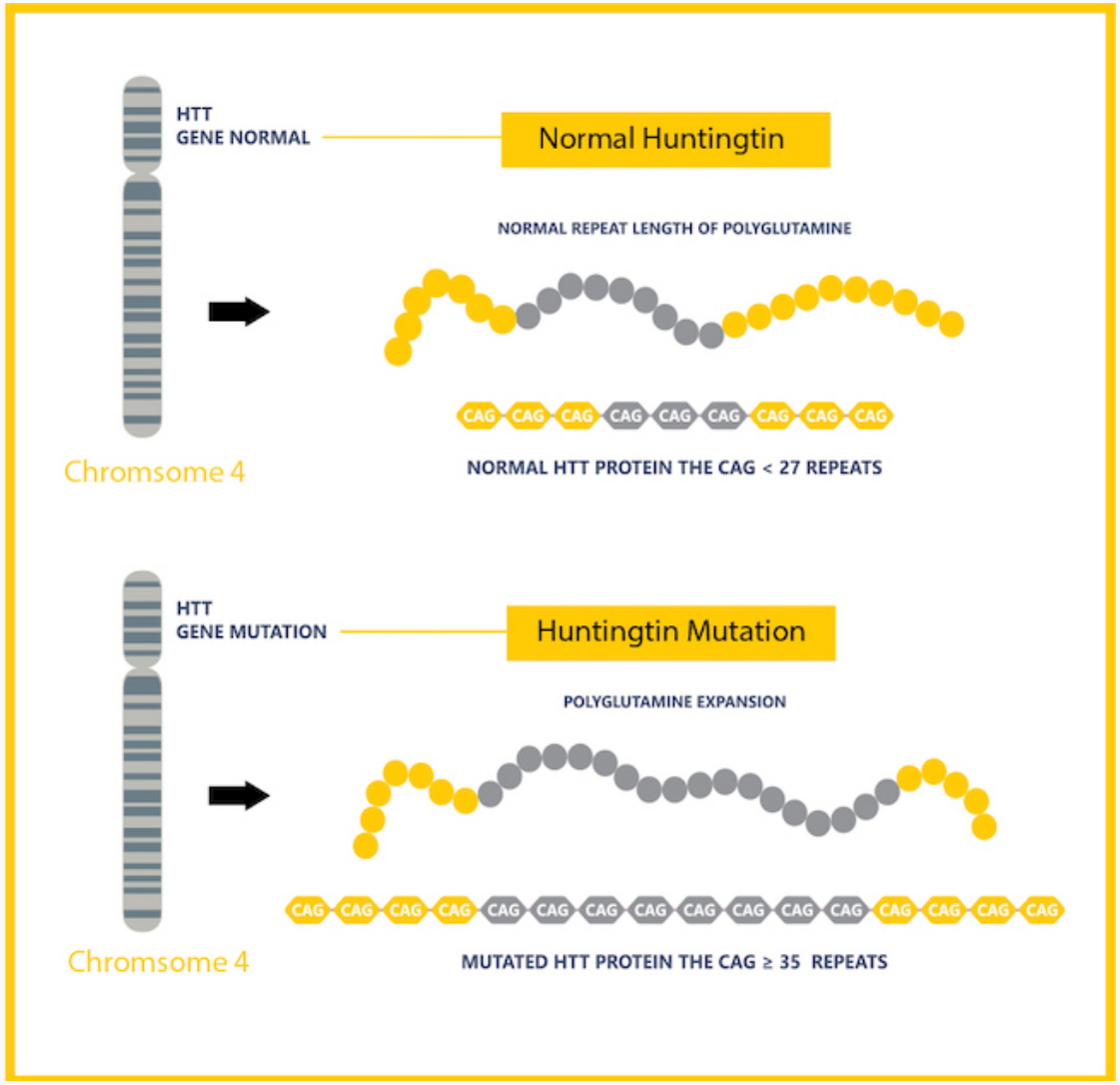
Objectives

- **Main Goal:** Improve phenotype prediction in HD using genomic language models (gLMs)
  1. Use of **Borzoi (gLM)** to predict tissue-specific expression from DNA
  2. Build multiple **XGBoost models** for classifying residual Age of Onset (AO)
  3. Analyze important **genes** & pathways from the features

Background

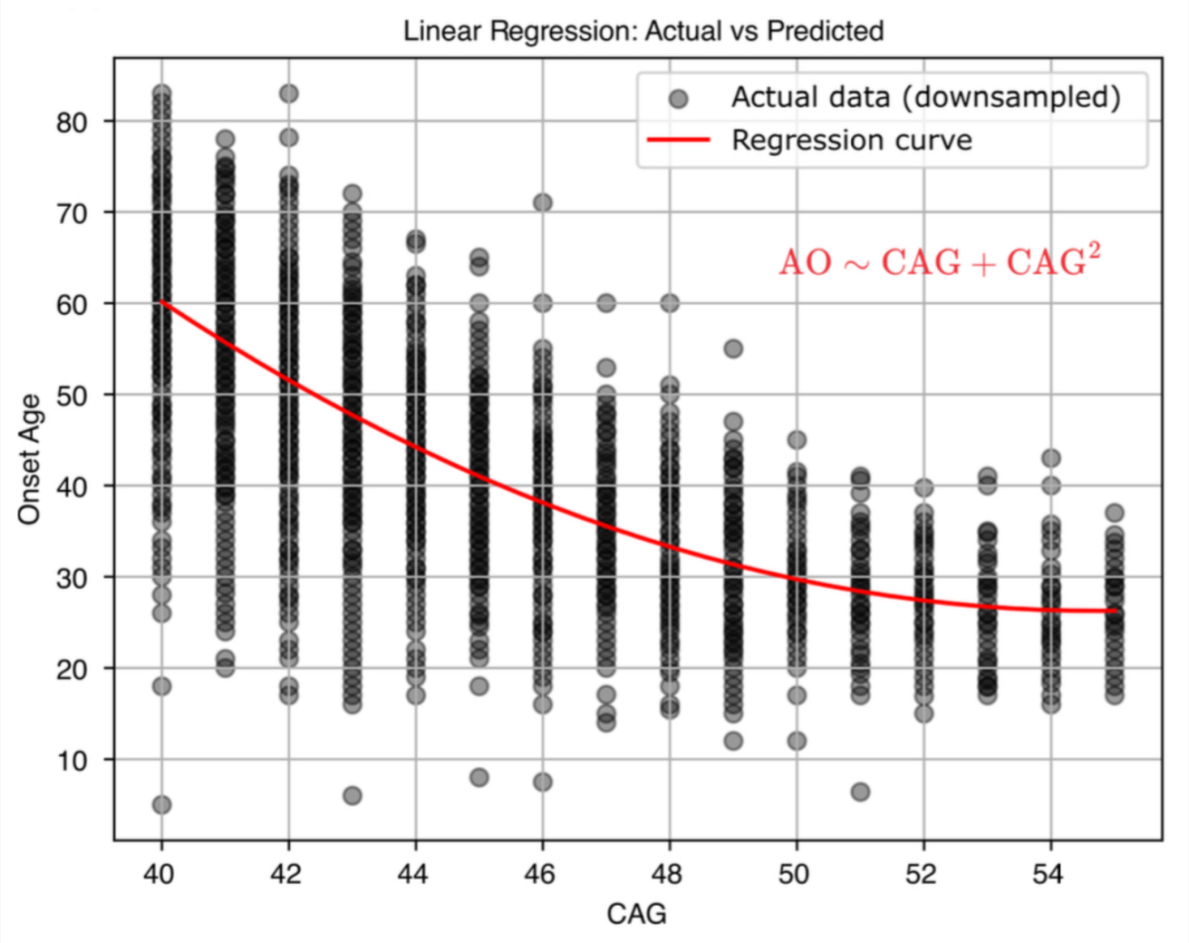
Background: Genetic Etiology in HD

Mutation of CAG repeats in the HTT gene



Background: Modeling the Effect of CAG Repeats on AO

Non-linear relationship between CAG repeats and AO

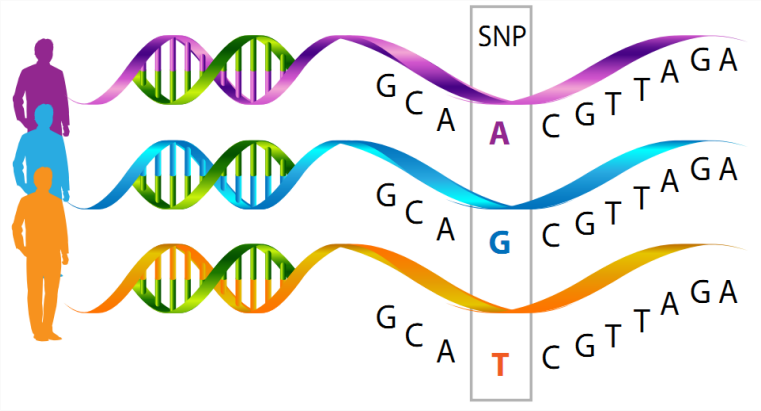


Background: Genetic Modifiers (GeMs) in HD

- CAG length explains 67% of onset variation
- Remaining 33% is from other **heritable** and environmental factors
  - 56% is heritable, meaning other **genes** influence onset timing
- **GeMs** are genes that change how **severe or early** the symptoms of HD are, even if people have the same CAG repeat length

Background: GWAS (Genome-Wide Association Studies)

- Observational study in genomics
- Searching for **small variations** in the entire genome such as single nucleotide polymorphisms **(SNPs)**



SNP variations

Background: GWAS (Genome-Wide Association Studies)

- **Phenotype Prediction**
  - GWAS helps **link DNA changes to symptoms**, showing that even normal CAG ranges and other genes beyond *HTT* can affect disease onset
- **Variant Effect Prediction (VEP)**
  - VEP predicts how **genetic variants** affect genes, proteins, or traits in organisms
  - In HD, VEP helps identify **modifier genes** beyond HTT
    - Identified *FAN1* as a gene that modifies HD AO

Background: Genomic Language Models (gLMs)

- LLMs trained on **DNA/RNA** sequences
- Learn **patterns** in genomic data (like NLP does with text)
- Capture functional elements
- Predict **variant effects** and **gene expression**
- Useful in **complex diseases** like HD

Background: State of the Art: Borzoi

- **Borzoi**
  - Supervised transformer trained on RNA-seq data
  - Predicts RNA coverage $\rightarrow$ **tissue-specific** gene expression
  - Mdels core RNA processing steps: transcription, splicing, and polyadenylation
  - Supports **VEP**
  - **Relevance to HD**
    - Captures non-coding effects of **modifier genes** like *FAN1*
    - Models expression in **brain-specific** contexts

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Background: Training and Performance of Borzoi

- Based on **Enformer**: CNN + Transformer + U-net
- Trained on RNA-seq from ENCODE, GTEx, and epigenetic data
- **Ensemble of 4 models** to improve generalization
- High predictive accuracy

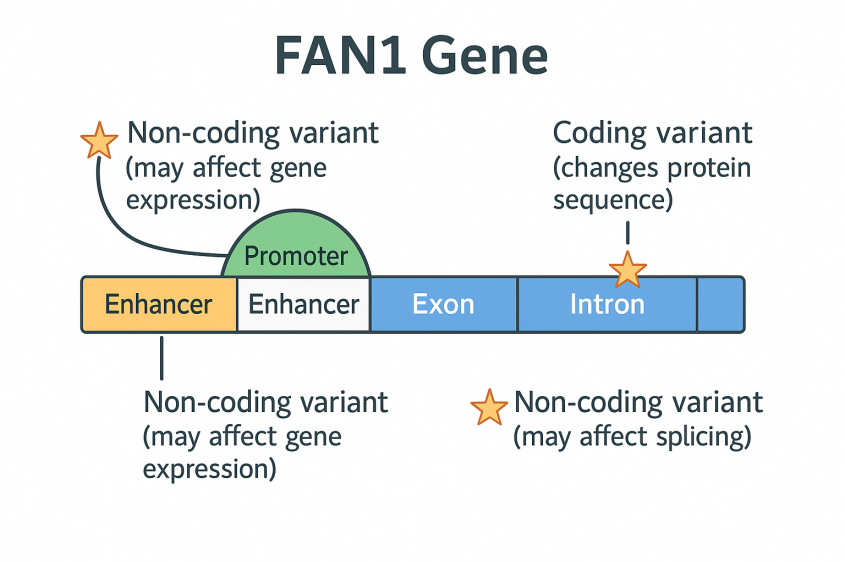
Methodology

Methodology: Genotype Data Sources

- Data from three major HD studies:
  - **Enroll-HD**
  - **GeM-HD**
  - **Registry**
- Total of **9,064 individuals**, each with:
  - SNP genotype data
  - CAG repeat length
  - Age of onset (AO)
- Focused on **common variants** (SNPs with frequency >1%)

Methodology: SNP Types Included

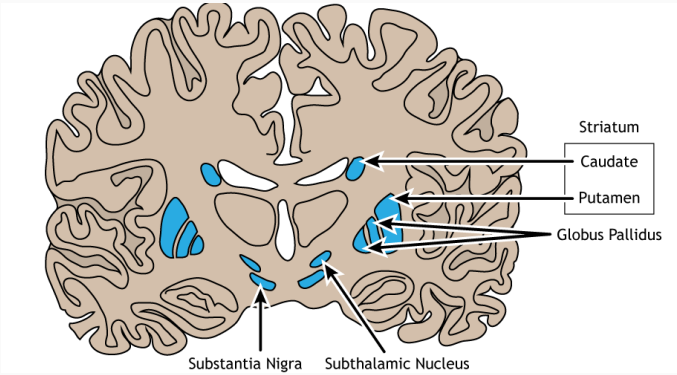
- Two types of SNPs included in the analysis:
  - **Protein-coding SNPs**:
    - 339,886 variants across 2,774 genes
  - **Regulatory SNPs**:
    - Enhancers
    - Promoters



FAN1 Variant

Methodology: Retrieval of RNA-seq Data

- **Data Source:**
  - RNA-seq data from healthy brain tissues (putamen and caudate)
  - Obtained from Sequence Read Archive (SRA)



- **Preprocessing Steps:**
  - Coverage files for each tissue were downloaded and merged to form aggregate signals
  - Converted the merged data into Borzoi-compatible `.w5` format
  - Created a target file (`targets_human.txt`) to guide brain-specific gene expression inference

Methodology: gLM Gene Expression Prediction

Use of Borzoi in This Thesis

- Applied a pretrained Borzoi model to predict RNA expression coverage for **caudate and putamen**
- Highlights impact of **SNPs in enhancers and promoters** that affect transcription
- Borzoi outputs log-transformed RNA-seq coverage values **("logSED")**, which estimate how much a genetic variant increases or decreases **gene expression** in a **specific tissue**
- Complements **genotype data** by adding functional interpretation for regulatory regions

Methodology: VCF Integration and Liftover

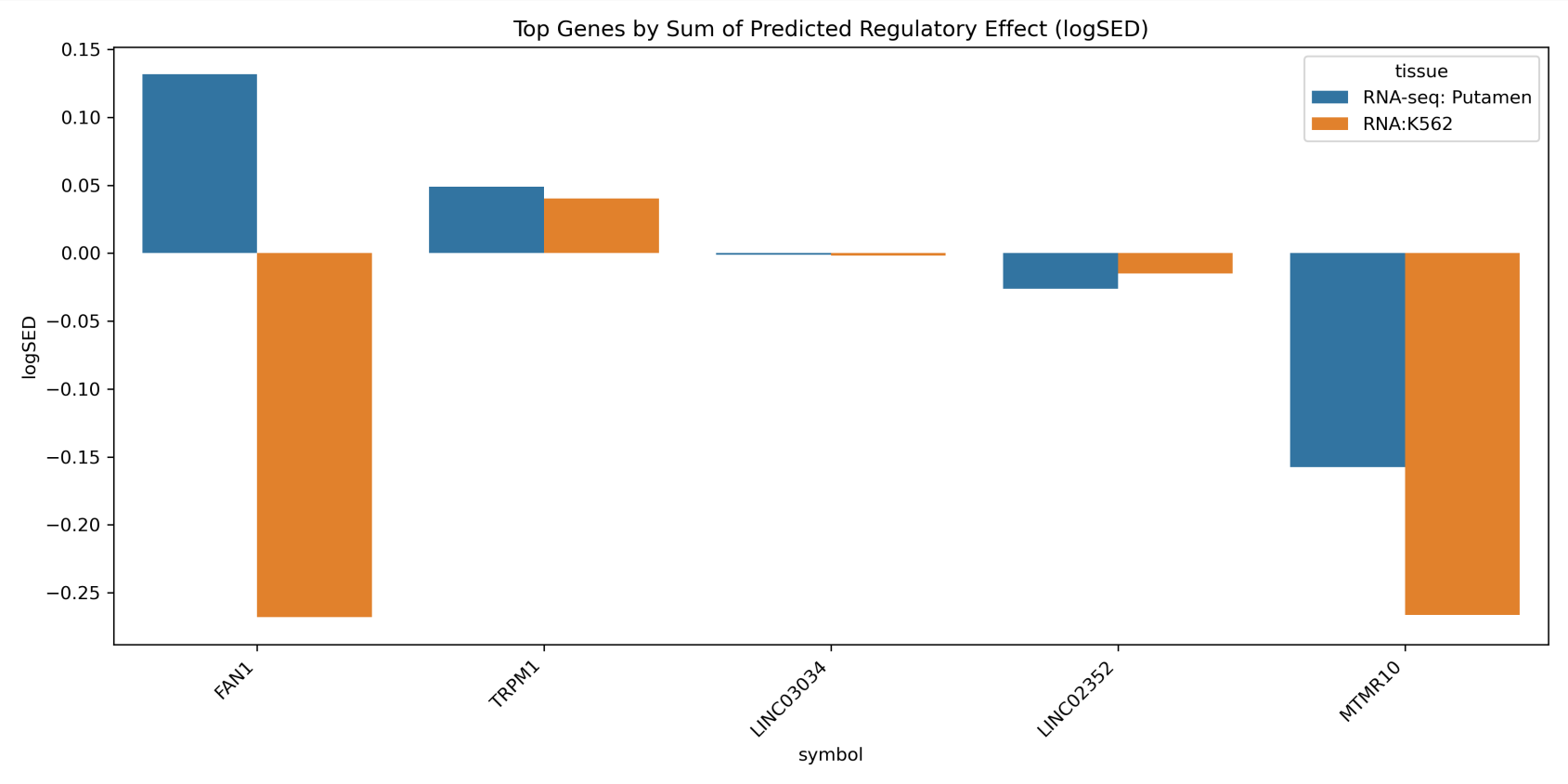
- VCFs = per-person genetic variants (SNPs)
- VCFs lifted from GRCh37 to GRCh38
- Borzoi combines reference genome with personal variants to model individual gene expression

Methodology: Borzoi Setup and Configuration

- Ran Borzoi with official test data (example from paper repository)
- Verified output format
- Ran tests to clarify meaningfulness of using tissue-relevant data

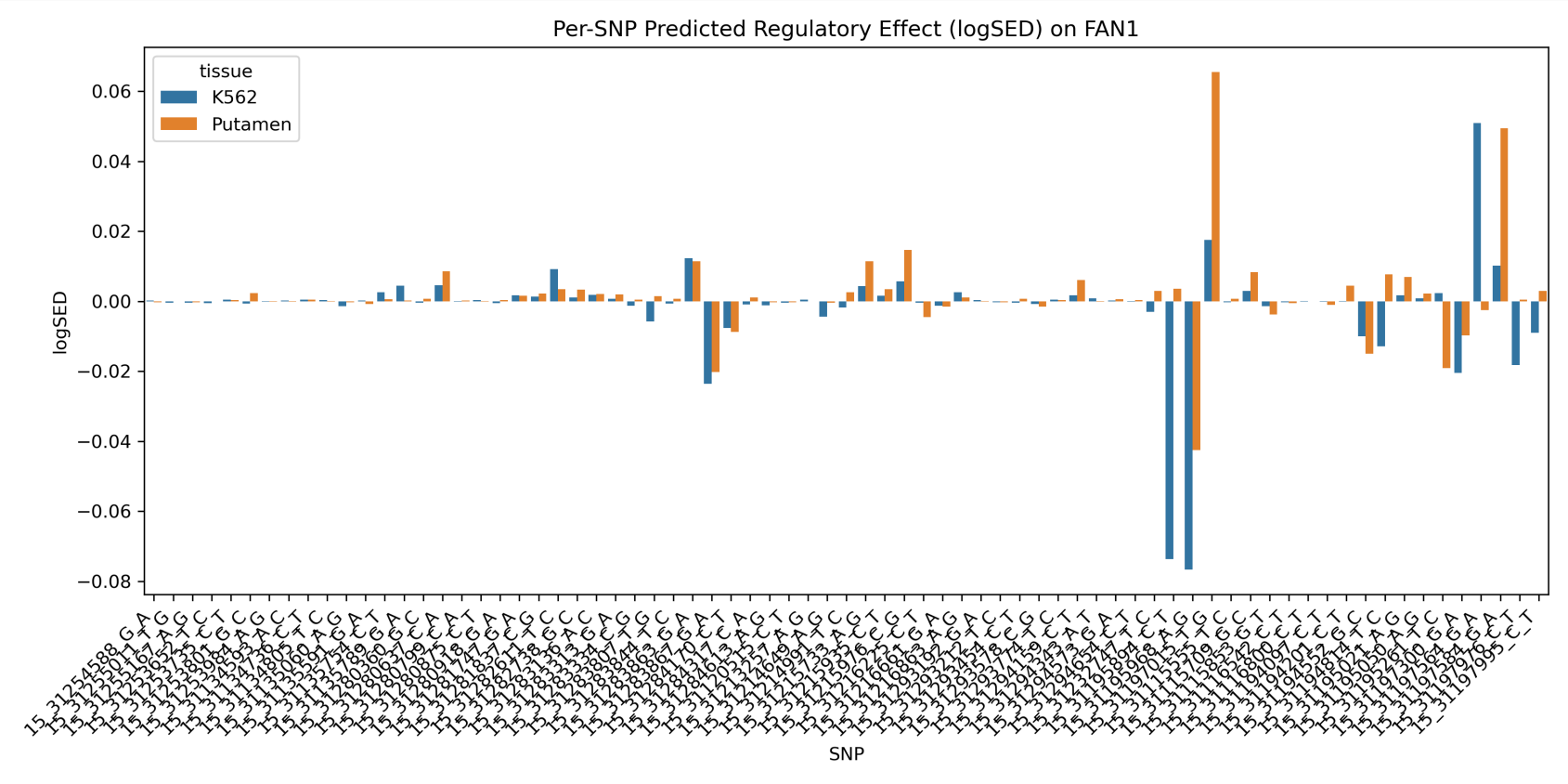
Methodology: Test 1

Comparison of Tissue-Specific Gene Regulatory Predictions



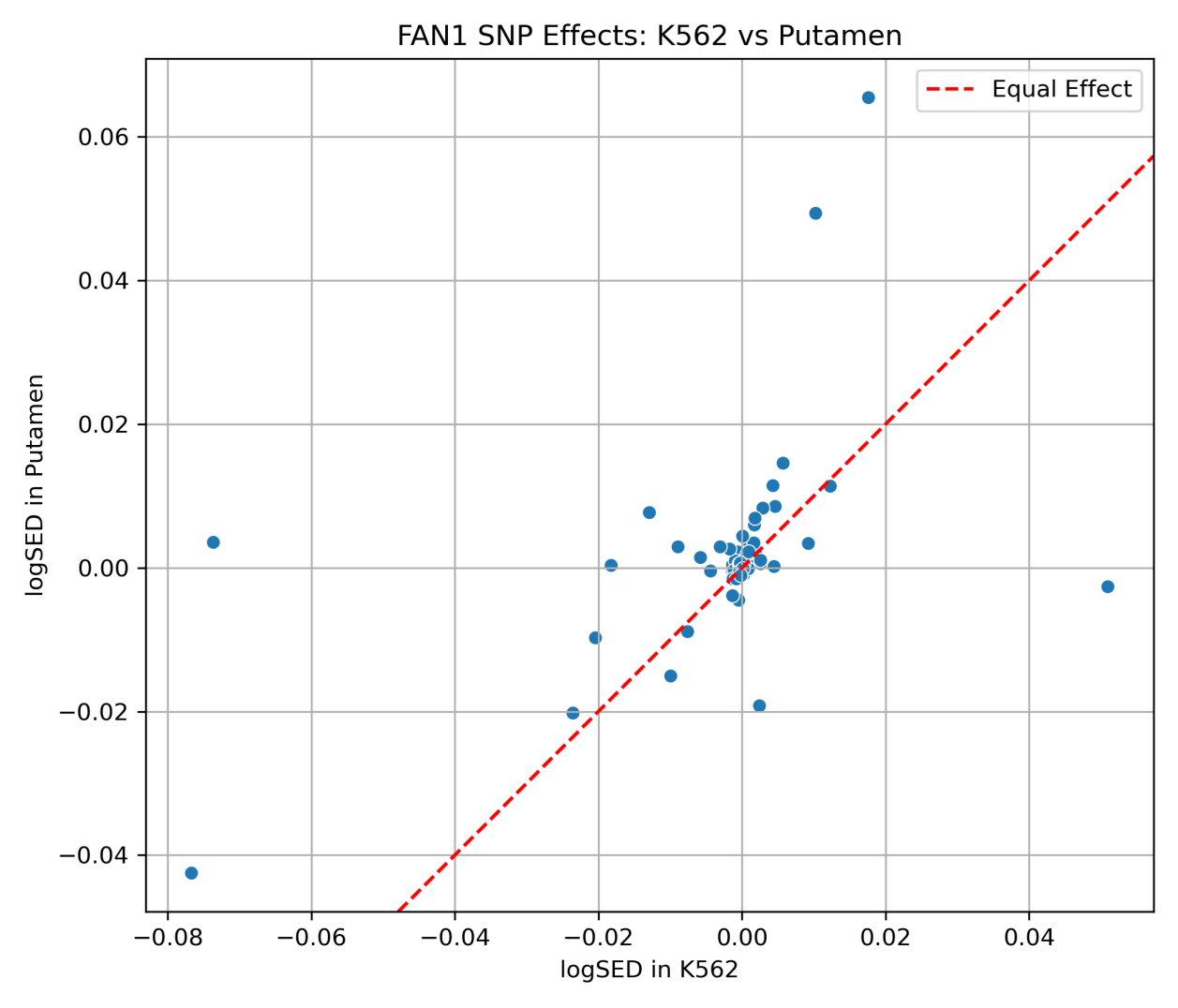
Methodology: Test 2

Per-SNP Regulatory Effects on FAN1 Across Tissues



Methodology: Test 3

Comparison of SNP Effects Across Tissues for FAN1



Methodology: Gaussian Weighting of LogSED values

Gaussian weights for SNPs based on enhancer distance

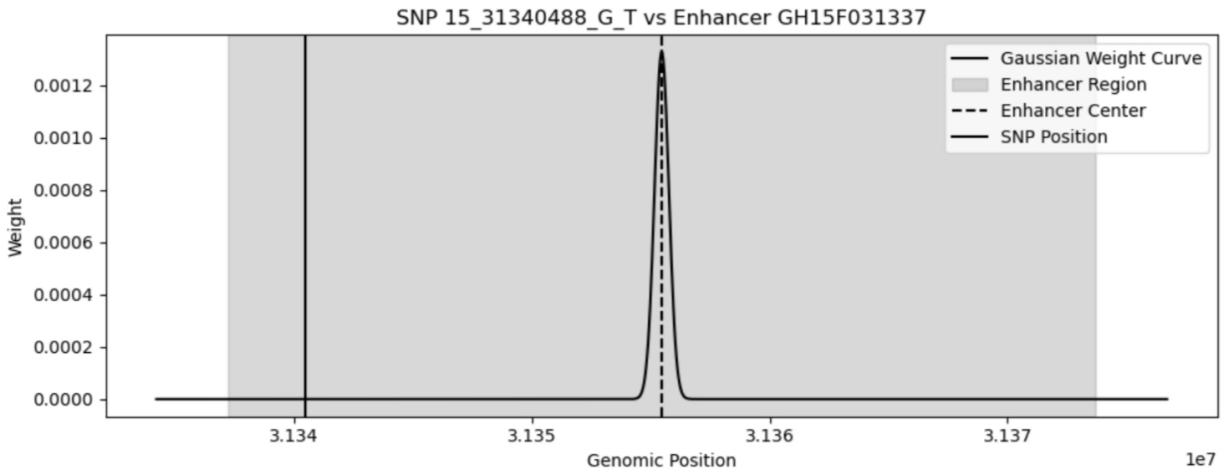
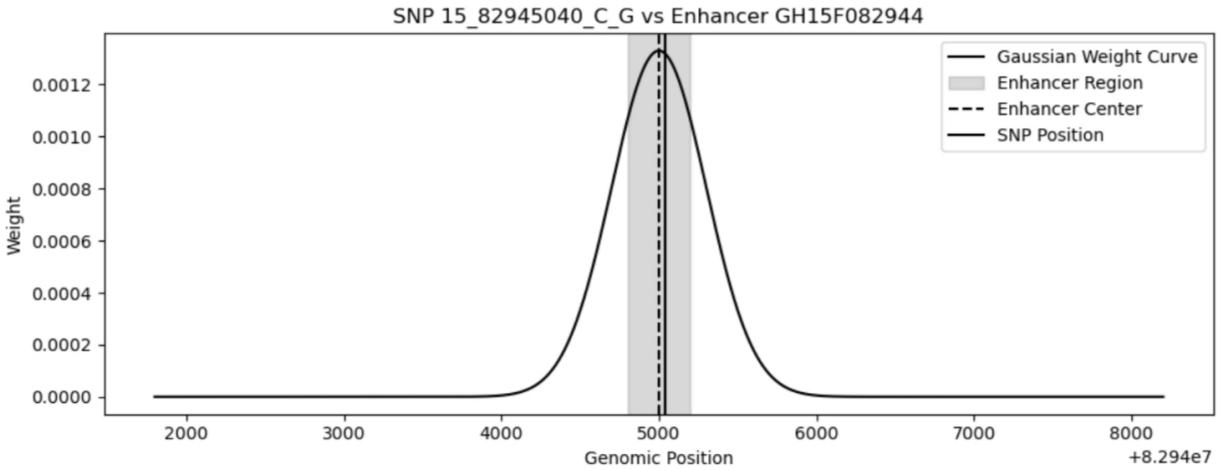
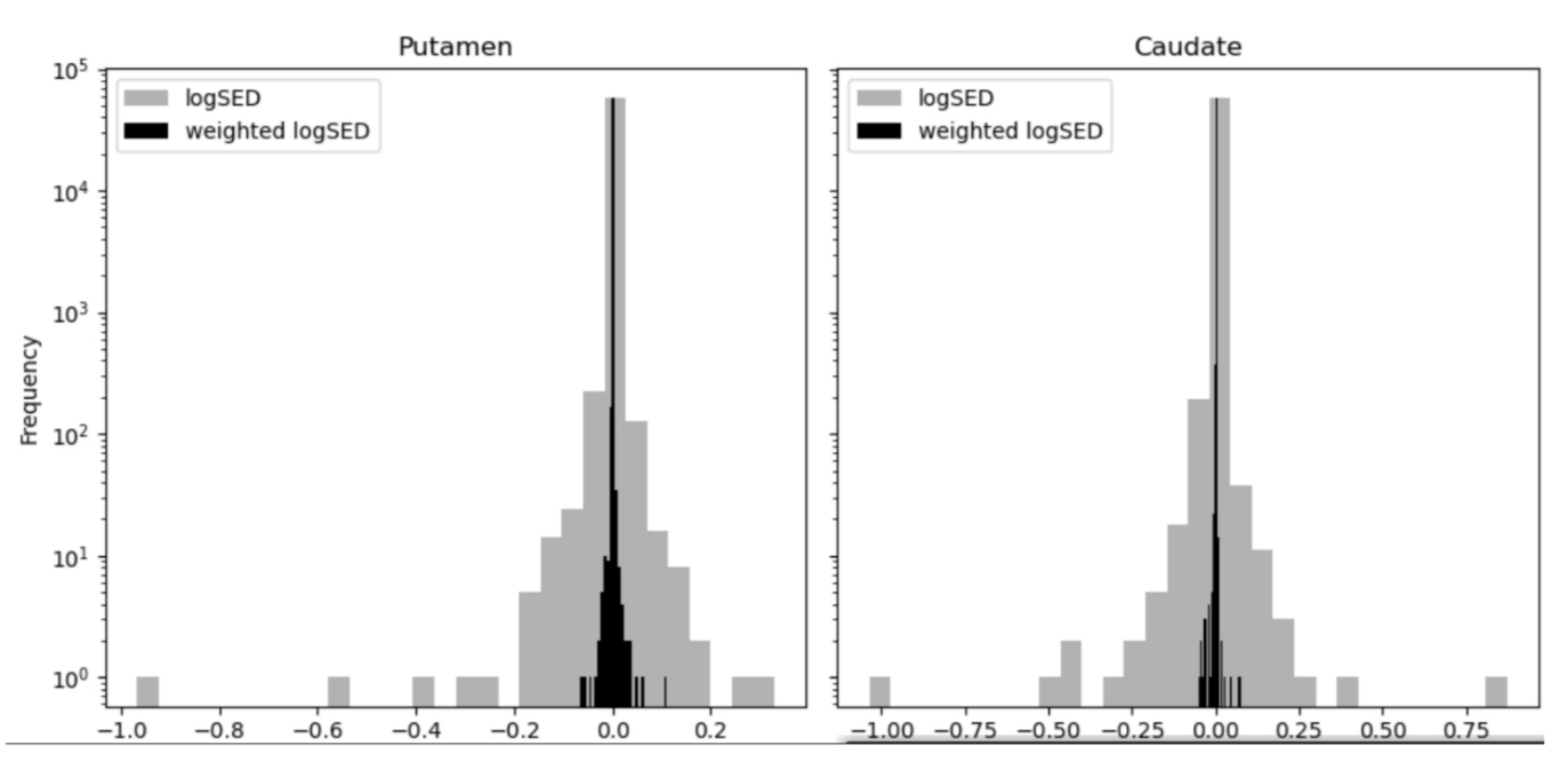


FIGURE 3.6: Gaussian Weighting of LogSED SNP1



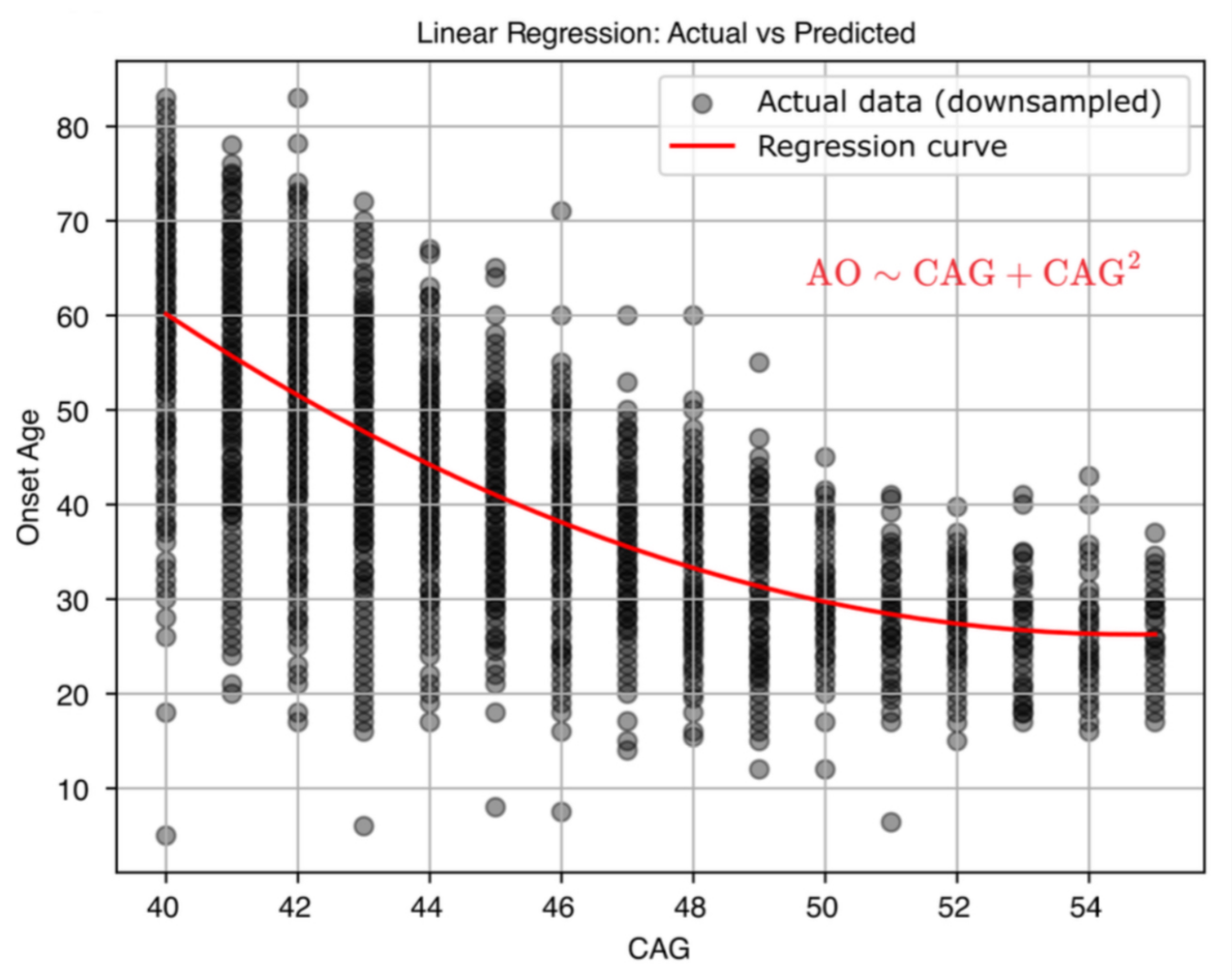
Gaussian Weighting of LogSED values

Before and After assigning Gaussian weights for SNPs



Methodology: Variant Selection and Modeling

To model genetic effects beyond CAG length, we used **Regression**



Methodology: Residual Computation and Binning

Compute Residuals

$$\textbf{Residual}(\textit{unexplainedbyCAG}) = \text{Observed AO} - \text{Predicted AO}$$

Binning Residuals

Divide residuals into 5 quantiles:

- **Class 0**: Much earlier than expected
- **Class 1**: Moderately earlier
- **Class 2**: About expected
- **Class 3**: Moderately later
- **Class 4**: Much later

Methodology: Phenotype Prediction Models with XGBoost

- **Goal:** Test if **predicted logSED** scores from Borzoi can improve phenotype prediction
- **Target:** Residual AO divided into 5 classes
- **XGBOOST**
  - Gradient-boosted decision tree model
  - Builds trees sequentially to reduce errors
  - Parameters tuned via grid search
  - Evaluated with **5-fold cross-validation**
  - Feature importance via gain score

Methodology: Phenotype Prediction Models with XGBoost

Note: All models included CAG repeat length and sex as covariates

Model 1: Baseline

- SNPs in protein-coding genes

Model 2: Expression-based (Borzoi)

- Borzoi logSED scores (Putamen & Caudate)
- Weighted by enhancer proximity, aggregated per gene

Model 3: Multimodal

- Combines genotype and Borzoi features

Methodology: Evaluation Metrics

- **Metric:** Balanced Accuracy (BA)
  - Average recall across all 5 classes (handles class imbalance)
- **Confidence Intervals:**
  - 90% CI via bootstrapping (1,000 resamples)
- **Statistical Tests:**
  - Binomial test, model performance exceeds random chance?
  - Wilcoxon signed-rank test (model comparison)

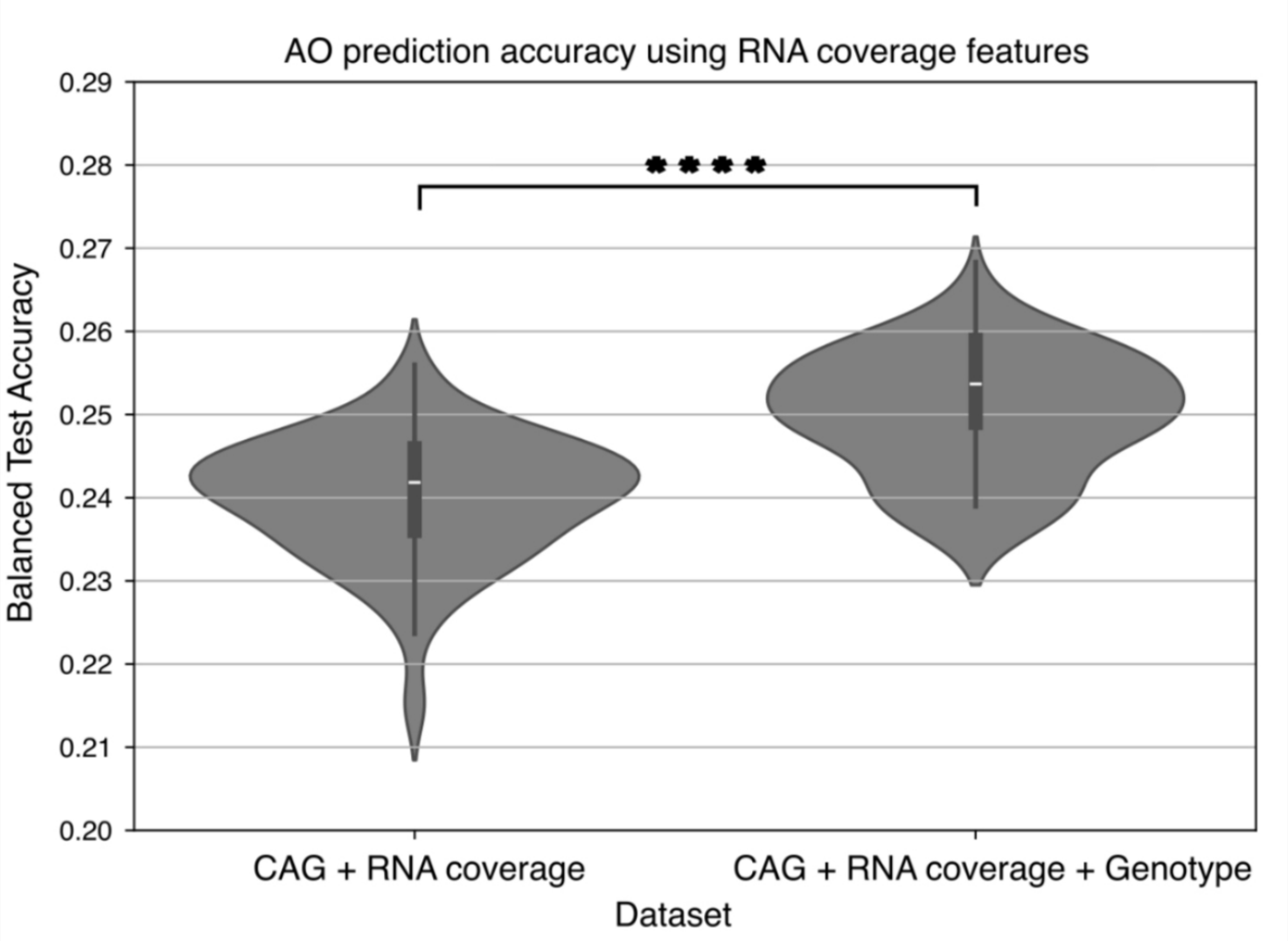
Results

Gene Expression Predictions (Expression-Only Model)

- **Median Balanced Accuracy (BA): 0.242**
  - Significantly better than chance (Binomial $p = 0.2$)
- **CAG repeat length** remained a strong predictor:
  - Used as first tree split in 24% of depth-2 models
  - Chosen in 65% of decision trees
- Identified 3 novel **enhancer-region variants** as candidate modifiers:
  - `19_50651485_A_C (rs1809186999)`
  - `5_60241142_G_A`
  - `1_157069597_G_A`

Results: Multimodal Vs Expression-only Model

Multimodal outperforms expression-only model



Results: Multimodal Phenotype Prediction Model

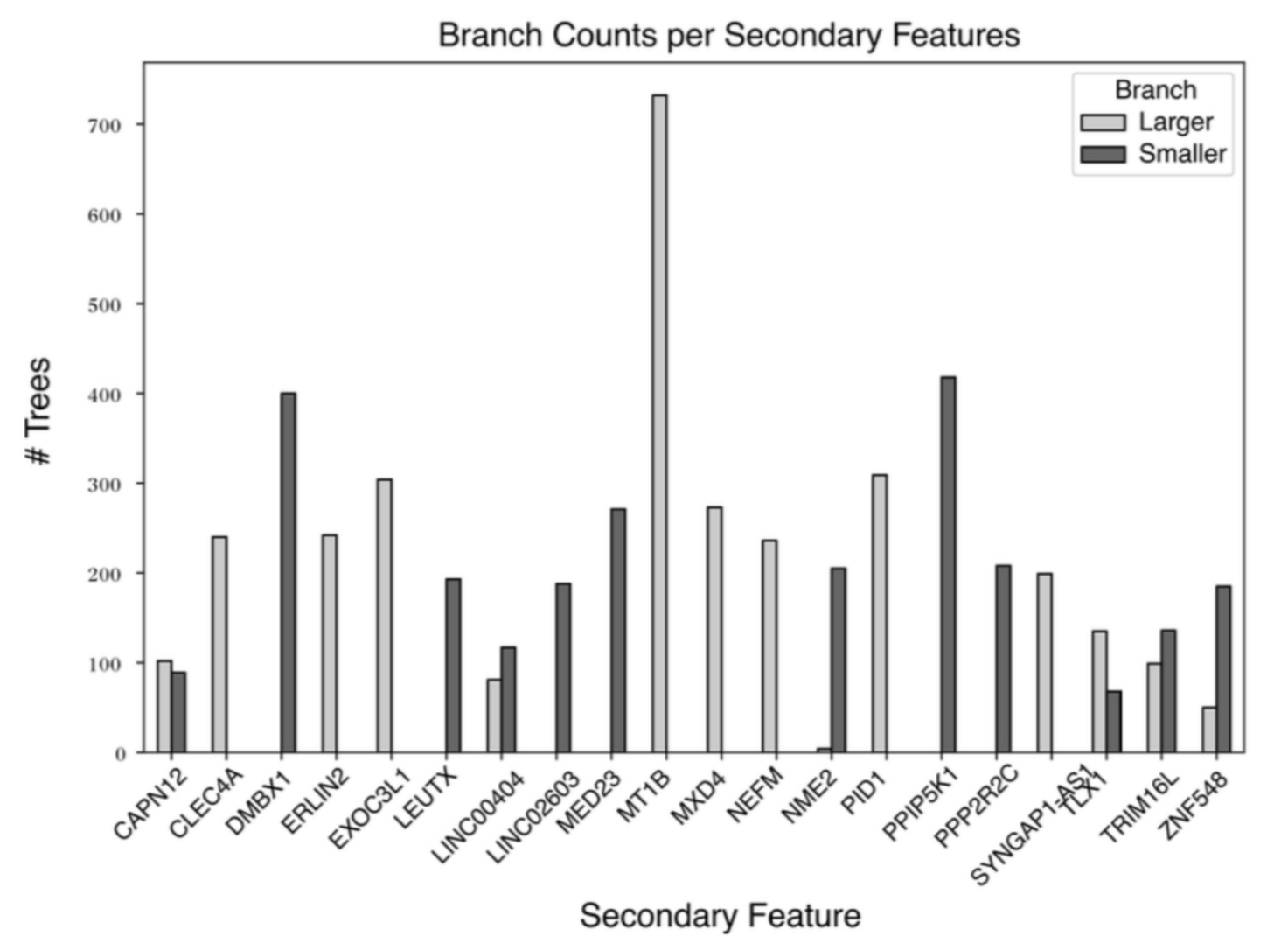
- No significant improvement over **genotype-only** ($p_{Wilc} = 0.19$)
- 44% of top 100 features were **expression-based** ($\downarrow$ to 35% after averaging by gene)
- Expression feature importance varies by **tissue**

Results: Biological Insights from Expression Features

- **Top expression features linked to core brain processes:**
  - DNA repair
  - gene regulation
  - Brain cell communication
- **Key genes highlighted by the model:**
  - **GRIK1:** Glutamate receptor which is important for brain signaling
  - **CUL2:** Helps break down damaged proteins
- **CAG repeat–specific effects:**
  - **MED23:** More important for individuals with shorter CAG repeats
  - **MT1B, DMBX1:** More important for longer CAG repeats

Multimodal Results

Frequency of Gene Features Used After CAG-Based Splits in Decision Trees



Conclusions

Conclusions

- Used Borzoi, a gLM, to predict **tissue-specific** gene expression in HD.
- Expression-only models **outperformed random**, capturing **regulatory signals** regions
- Multimodal model matched genotype-only performance but revealed **additional modifiers**
- Identified **biological processes** linked to HD (transcription regulation, DNA binding, and glutamate signaling)
- Found **CAG-dependent modifier effects**, suggesting length-specific regulation

Limitations and Future Work

- **Enhancer-gene mappings** from GeneHancer may be incomplete
- Larger HD cohorts may be needed to detect weaker genetic modifier effects
- Future models could support **in silico perturbation** experiments
  - Simulate how changing gene expression may delay symptom onset
  - Identify potential targets for gene therapy

Thank You

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